Bachelor's Degree in Techniques for Software Development

Guide for new students

Universitat Oberta de Catalunya

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Bachelor's Degree

Techniques for Software Development

Guide for new students

Introduction

This guide offers you all the key information you need to begin the bachelor's degree programme.

Throughout your studies, whenever you feel the need for some advice or guidance, you can always turn to your tutor. Contact them at any time by email or via the tutor's classroom on the Campus.

The **tutors** are:

- First-year tutor: **Jordi Pueyo (jpueyob@uoc.edu)**
- Follow on tutor: **Eduard Capell (ecapell@uoc.edu)**

Presentation

Today's world has many automated processes, with intelligent and interconnected new devices opening doors to novel ways of interacting with our surroundings. Thanks to software we are able to control these activities and devices, making the most of their potential.

This has created high levels of demand for software developers; indeed, this role is among those that companies worldwide are keenest to recruit.

In this context, the Bachelor's Degree in Techniques for Software Development will provide students with expertise to **develop** software, applications and services for any field. It will also train them as **administrators** of the devices, networks, and systems in which the applications are used.

Students will learn the **software engineering techniques** used to build programs as well as the **programming languages**, tools and technologies most commonly used to develop software for different platforms. Furthermore, they will develop their knowledge of **web programming and standards**, **databases**, **human-computer interactions**, **networks** and **operating systems**, all of which are key factors in the creation and administration of usable, efficient and scalable apps.

Language

The language of instruction for this bachelor's degree is **English**. This is the language that is used in communications with teaching staff, in learning resources and in the continuous assessment process.

This use of English is one of the programme's strong points, as it will sharpen your abilities in a language that is essential in the world of programming, preparing you for a job market that is increasingly globalized.

For admission to the programme you must provide proof of **English level B2** (as per the Common European Framework of Reference for Languages).

Enrolment calendar and start of the academic year

Course enrolments and the teaching of courses are based on a calendar of two semesters per year:

1. From September to January
2. From March to July

Key dates

Start of the semester: 27 September 2023
End of the semester: 9 February 2024

Programme of study

This bachelor's degree comprises **8 basic courses**, **16 compulsory courses**, **8 optional courses** (from which you will choose 4 or, if you choose to do an internship, 3) and a **final project**. In total, you must pass **180 ECTS credits**.

BASIC COURSES			
	ECTS credits *	First offered	Semester taught
Algebra	6	September 2021	Both
Logic	6	September 2021	Both
Web-Based Teamwork	6	September 2021	Both
Fundamentals of Programming	6	September 2021	Both
Programming in Practice	6	September 2021	Both
Web Programming	6	September 2021	Both
Software Engineering	6	September 2021	Both
Computer Structure	6	September 2021	Both
COMPULSORY COURSES			
	ECTS credits *	First offered	Semester taught
Entrepreneurship	6	February 2022	Both
Object-Oriented Programming	6	September 2022	Both
Data Structures	6	September 2022	Both
Web Standards and Languages	6	September 2021	Both
Advanced Web Programming	6	February 2022	Both
Software Design Patterns	6	February 2022	Both
Software Architecture	6	February 2022	Both
Human-Computer Interaction	6	February 2022	Both
Introduction to Databases	6	September 2022	Both
Database Design	6	March 2023	Both
Operating Systems	6	September 2022	Both
Networks and Internet Applications	6	September 2022	Both
Network and System Administration	6	March 2023	Both
Cloud Computing	6	September 2022	Both
Mobile App Development	6	September 2022	Both

Business and IT Management	6	September 2022	Both
Final Project	12	March 2023	Both
OPTIONAL COURSES			
	ECTS credits *	First offered	Semester taught
Communication Skills for ICT Professionals	6	March 2023	March to July
Distributed Systems	6	March 2023	March to July
Security in Computer Networks	6	March 2023	March to July
E-commerce	6	September 2022	September to February
Fundamentals of Computers	6	March 2023	March to July
Embedded Systems	6	September 2023	September to February
Fundamentals of Information Systems	6	September 2023	September to February
Internship	12	March 2023	Both

* Each credit represents 25 hours' study or work.

Teaching methods and assessment

For each course you'll have a course instructor who will guide your learning process and provide answers to any queries. In the online classrooms there are various communication channels, used to set tasks which will help you keep up-to-date throughout the course and to share information with fellow students.

To pass the courses you must complete continuous assessment activities. These include tasks such as practical work, debates, giving online presentations, answering tests, or preparing projects or deliverables. The learning resources required for these activities (teaching materials, information sources, support tools, etc.) will be found in the classroom.

Detailed and up-to-date information on each course (general details, course description, contents, materials and assessment model) will be found in the **course plan**. The course plans can be found on the Campus:
More UOC > Programmes of Study > Bachelor's Degree in Techniques for Software Application Development > Courses.

Assessment model	
Algebra	CA+EX or EX
Computer Structure	**CA**
Fundamentals of Programming	**CA**
Logic	CA+ST or EX
Software Engineering	**CA+EX**
Programming in Practice	**CA+ST**
Web Programming	**CA**
Web-Based Teamwork	**CA**
Network and Systems Administration	**CA**
Software Design Patterns	**CA**
Software Architecture	**CA**
Cloud Computing	CA
Mobile App Development	CA
Database Design	**CA**
Data Structures	**CA**
Object-Oriented Programming	**CA**
Business and IT Management	CA
Entrepreneurship	**CA**
Human-Computer Interaction	**CA**
Web Standards and Languages	**CA**
Advanced Web Programming	**CA**
Operating Systems	**CA**
Final Project	CA
Introduction to Databases	**CA+EX**
Network and Internet Applications	**CA**
E-Commerce	CA
Communication Skills for ICT Professionals	AWAITING DECISION
Fundamentals of Computers	AWAITING DECISION
Fundamentals of Information Systems	CA
Internship	CA
Security in Computer Networks	CA
Distributed Systems	AWAITING DECISION
Embedded Systems	**CA+ST**

- CA = Continuous assessment.
- ST = Synthesis test *
- EX = Exam

* Assessment of whether the course's objectives have been accomplished, its competencies developed and its contents learned. To take it, you need to have passed the continuous assessment. It will last no longer than one hour.

Enrolment planning

This bachelor's degree programme can be studied full-time over **three academic years (six semesters)** or part-time over **six years (twelve semesters)**.

However, there is no time limit for finishing the programme; each student can adjust the length and pace of their studies to fit in with their other commitments and the time they have available.

Therefore, it's important to plan the courses that you enrol on each semester carefully, taking into account the time you'll be able to dedicate to them.

a) Enrolment recommendations and general recommendations

- Your tutor will provide you with personalized guidance to help you plan your course enrolments based on your needs, interests, availability and existing knowledge.
- Some courses may be recognised due to your prior studies or professional experience (see the Credit recognition section of this guide).
- Part-time students usually enrol on two or three courses, whereas full-time students typically enrol on five.
- For the first semester of the programme, we recommend choosing one of these two course packages and only enrolling courses from that package:

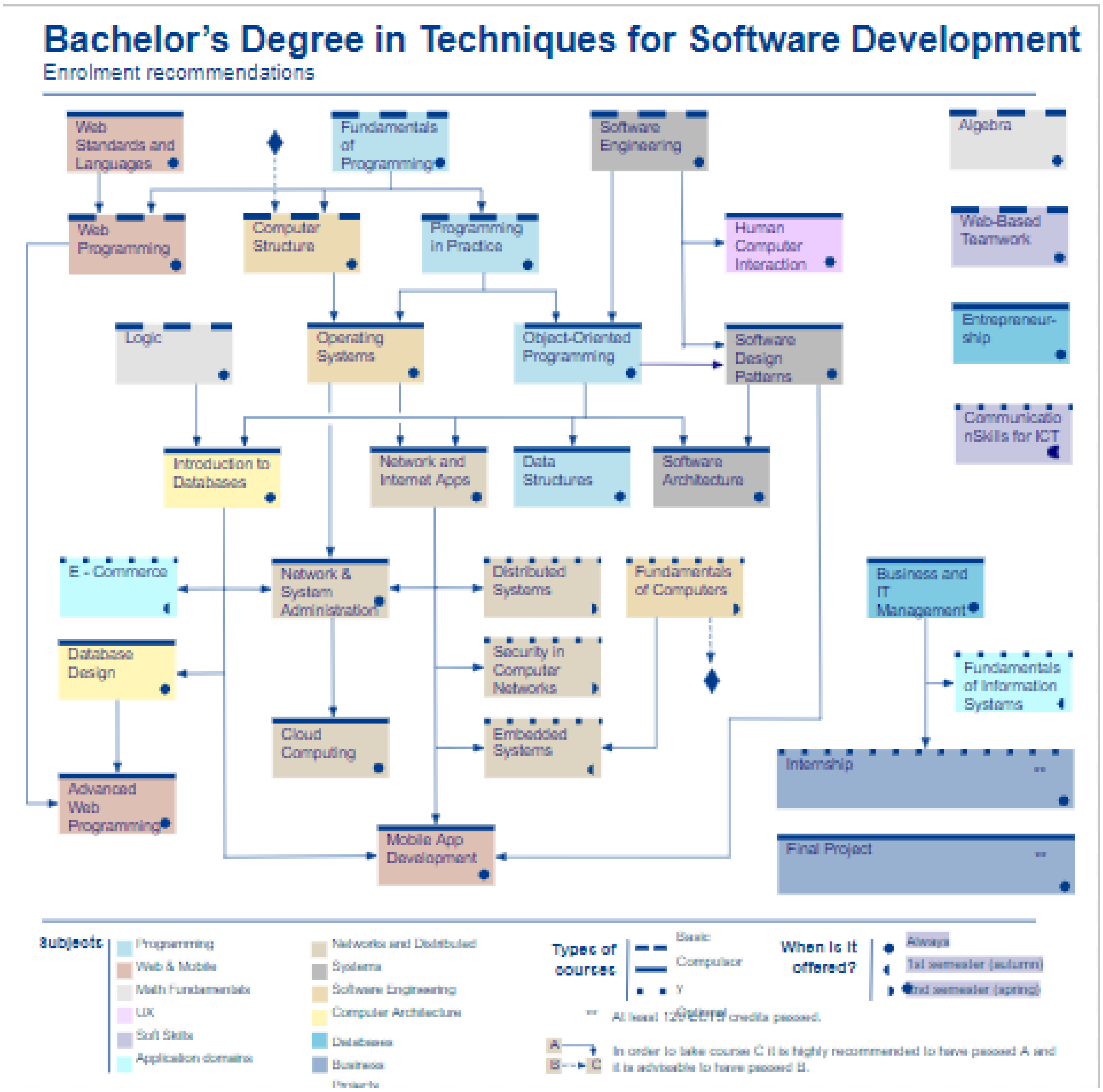
  - **Package 1 - Software Development:** Fundamentals of Programming, Software Engineering, Web Standards and Languages
  - **Package 2 - Fundamentals of Mathematics:** Algebra, Business and IT Management, Web-Based Teamwork

  These packages have been specially designed for your first enrolments. You don't have to do all three courses from the package during the same semester; you can do just one or two. If you want to do two, we recommend the ones that have been underlined above.

b) Course enrolment recommendations diagram

The diagram below shows which courses build upon content learned in other courses and, because of this, should be taken in a set order. If there is an arrow leading from course X to course Y, it means that you need some of the content learned in course X before you can do course Y. Therefore, these are 'strong' recommendations to do one course before the other.

It is possible to enrol on courses without following these recommendations (for example, you might do this based on your professional experience or existing knowledge). However, in general you are expected to follow these guidelines. If you have any doubts, your tutor can provide advice.



[Click the following link to access a larger version of this diagram (PDF format).]

Credit recognition for prior studies and professional experience

If you have professional experience or have completed other studies (partially or fully) that relate to the content of this bachelor's programme, demonstrating evidence of this may provide you with credits for this bachelor's degree.

- **Prior studies**: if you have done advanced-level vocational training in

Spain **(CFGS)** or studied on another bachelor's degree (completed or not), you can request credit transfers for courses you have already passed.

- **Professional experience**: if your experience clearly relates to the content covered in this bachelor's programme, you can obtain credit recognition for up to 15% of the 180 credits. That works out as a maximum of four courses (or three courses if one of them is the Internship course).

You can check all the details and the calendar for requesting credit recognitions in the Virtual Campus sections *Procedures / Recognitions and validations / Request prior studies assessment* and *Procedures / Recognitions and validations / Request the recognition of professional experience*.

If your credit recognition request is validated, you must then enrol on the courses before completing the programme in order for them to be included in your academic record. The price of courses passed by means of credit recognition is lower than the normal price.

Enrolment specifications: Internship and Final Project courses

Typically, people studying the programme full-time would enrol on the Internship course (which is optional) in the fifth semester and the Final Project course in the sixth semester.

It is important to know that these two courses **cannot be taken until you have passed at least 120 ECTS credits**.

In general, very few exceptions can be made to this rule; any such special cases would **have to be discussed first with the tutor and would then require authorization from the programme director**.

For the Final Project course, you will have to choose a field of study in which to do your project. In some cases, the teaching staff will propose fields of study; in other cases, you can come up with your own.

In any case, we recommend that each student should contact their tutor before enrolling on the Final Project course so that they can receive advice on fields of interest before making their decision.

You will receive more information on the enrolment process, the process for choosing where to do the internship and the focus of your final project in the future.

Teaching figures

Tutor

Your tutor is the first person to turn to if you have any doubts or questions about your studies. They will provide you with guidance and advice throughout your time studying this programme. Contact them at any time by email or via the tutor's classroom on the Campus. You will have a **First year tutor** who'll guide you through your firsts semesters, and then switch to a **Follow-on tutor** who will stay with you through the rest of the programme.

Coordinating professors and course instructors

Coordinating professors organize and integrate each course's contents, focusing on students' learning processes. They contribute expertise in the field of study and are responsible for coordinating the course instructors.

For each course you will have a course instructor, and this is the person who you will encounter on a day-to-day basis for matters relating to the course (the learning process, answering questions, and messages in the forums or directly via email). As such, the course instructor is the main teaching figure who will provide you with a personalized learning experience as well as continuous and final assessment for each course.

Programme director

The programme director is in charge of the programme as a whole. Their academic responsibilities include managing the courses' coordinating professors and ensuring that the resources and teaching quality are up to standard.

Student Help Service

If you have any queries or problems with academic or administrative procedures, you should go to the **Help Service** (found at the top of the Virtual Campus).

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